

Problem #1

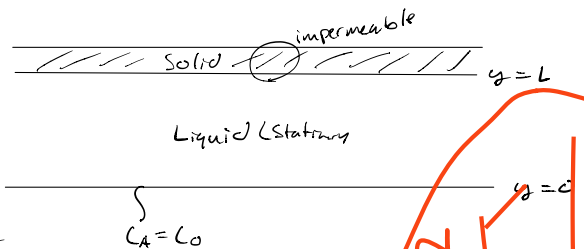
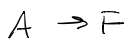
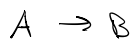
Assume

Stationary liquid film

Irreversible

Isothermal

(dimensionless) steady state



Rayan Saig
CHE312
Exam #3

$$N_{Ax} = -C_{AB} \frac{\partial x_A}{\partial y} + \text{Both } 0 \text{ bc stationary}$$

$$N_{Ax} = -D_{AB} \frac{\partial C_A}{\partial y}$$

From mass balance

$$(N_{Ay}|_{y=0} - N_{Ay}|_{y=L} + \dot{N}_{SOF} = 0) \leftarrow \begin{matrix} \div S_{Oy} \\ \uparrow \\ \text{all rxns} \end{matrix}$$

$$-\frac{\partial N_{Ay}}{\partial y} + \dot{N} = 0$$

$$-\frac{\partial N_{Ay}}{\partial y} - k_B - k_C - k_F C_A = 0$$

$$-\frac{\partial}{\partial y} (-D_{AB} \frac{\partial C_A}{\partial y}) - k_B - k_C - k_F C_A = 0$$

$$D_{AB} \frac{\partial}{\partial y} (\frac{\partial C_A}{\partial y}) - (k_B + k_C + k_F C_A) = 0$$

$$\frac{\partial}{\partial y} (\frac{\partial C_A}{\partial y}) - \frac{(k_B + k_C + k_F C_A)}{D_{AB}} = 0$$

$$\frac{\partial^2 C_A}{\partial y^2} - \frac{k_B + k_C + k_F C_A}{D_{AB}} = 0$$

$$\frac{\partial^2 C_A}{\partial y^2} - \frac{k_F}{D_{AB}} C_A - \frac{k_B + k_C}{D_{AB}} = 0$$

$$\frac{\partial^2 C_A}{\partial y^2} - \frac{k_F}{D_{AB}} C_A = \frac{k_B + k_C}{D_{AB}}$$

$$\sqrt{\frac{k_F}{D_{AB}}} y - \sqrt{\frac{k_F}{D_{AB}}} y$$

$$C_A(y) = C_1 e^{\sqrt{\frac{k_F}{D_{AB}}} y} + C_2 e^{-\sqrt{\frac{k_F}{D_{AB}}} y} - \frac{k_B + k_C}{k_F}$$

BC#1 when $y=0$

$$C_0 = C_1 + C_2 - \frac{(k_B + k_C)}{k_F}$$

$$C_1 = C_0 - C_2 + \frac{k_B + k_C}{k_F}$$

$$C_1 = C_0 - C_2 e^{\sqrt{\frac{k_F}{D_{AB}}} L} + \frac{k_B + k_C}{k_F}$$

$$C_1 (1 + e^{\sqrt{\frac{k_F}{D_{AB}}} L}) = C_0 + \frac{k_B + k_C}{k_F}$$

$$C_1 = \frac{C_0 + \frac{k_B + k_C}{k_F}}{(1 + e^{\sqrt{\frac{k_F}{D_{AB}}} L})}$$

BC#2

$$\frac{\partial C_A}{\partial y} \bigg|_{y=L} = C_1 \sqrt{\frac{k_F}{D_{AB}}} e^{\sqrt{\frac{k_F}{D_{AB}}} L} - C_2 \sqrt{\frac{k_F}{D_{AB}}} e^{-\sqrt{\frac{k_F}{D_{AB}}} L} + 0$$

$$0 = C_1 \sqrt{\frac{k_F}{D_{AB}}} e^{\sqrt{\frac{k_F}{D_{AB}}} L} - C_2 \sqrt{\frac{k_F}{D_{AB}}} e^{-\sqrt{\frac{k_F}{D_{AB}}} L}$$

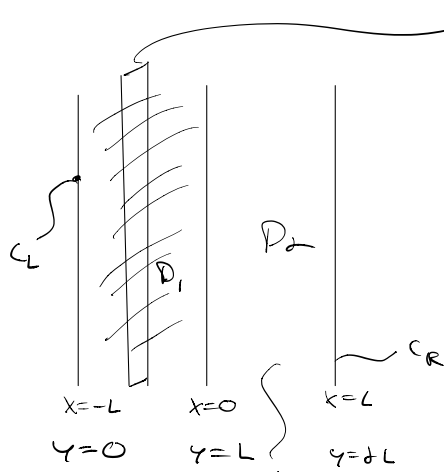
$$C_2 = \frac{\sqrt{\frac{k_F}{D_{AB}}}}{e^{\sqrt{\frac{k_F}{D_{AB}}} L}} = C_1 \sqrt{\frac{k_F}{D_{AB}}} e^{\sqrt{\frac{k_F}{D_{AB}}} L}$$

$$C_2 = C_1 e^{\sqrt{\frac{k_F}{D_{AB}}} L}$$

$$C_A(y) = \left(\frac{C_0 + \frac{k_B + k_C}{k_F}}{1 + e^{\sqrt{\frac{k_F}{D_{AB}}} L}} \right) e^{\sqrt{\frac{k_F}{D_{AB}}} y} + \left(\frac{C_0 + \frac{k_B + k_C}{k_F}}{1 + e^{\sqrt{\frac{k_F}{D_{AB}}} L}} \right) e^{-\sqrt{\frac{k_F}{D_{AB}}} y} - \frac{k_B + k_C}{k_F}$$

Problem #2

Steady state
1 dimension



increment
range $-L \rightarrow L$

when $x=0$ homogeneous
component other order

$$S(k_0)$$

$$\frac{\partial}{\partial x} - D_{AB} \frac{\partial C_A}{\partial x} + k_0(x-L) = 0$$

$$x(x+L)$$

$$\frac{\partial}{\partial x} \frac{\partial C_A}{\partial x} + k_0 = 0$$

only from D_2 region

region #1

$$C_{A1}(x) = C_1 x + C_2 \quad \text{BC#1}$$

$$C_L = -C_1 L + C_2$$

$$C_2 = C_L + C_1 L$$

Region #2

$$C_{A2}(x) = \frac{-k_0}{2D_2} x^2 + C_3 x + C_4$$

BC#2

$$C_R = \frac{-k_0 L^2}{2D_2} + C_3 L + C_4$$

$$C_4 = C_R + \frac{k_0 L^2}{2D_2} - C_3 L$$

$$C_{A2}(x) = \frac{-k_0}{2D_2} (x^2 - L^2) + C_3 (x-L) + C_R$$

$$\text{BC#1} \quad x = -L \quad C_A = C_L$$

$$\text{BC#2} \quad x = L \quad C_A = C_R$$

$$\text{BC#3} \quad x = 0 \quad -D_1 \frac{\partial C_{A1}}{\partial x} \bigg|_{x=0} = -D_2 \frac{\partial C_{A2}}{\partial x} \bigg|_{x=0}$$

$$\text{BC#4} \quad \phi = \frac{C_{A1}(x=0)}{C_{A2}(x=0)}$$

assume $\phi = 1$

$$\text{eval} \quad \text{BC#3} \quad -D_1 \frac{\partial C_{A1}}{\partial x} = -D_2 \frac{\partial C_{A2}}{\partial x}$$

$$D_1 (C_1) = D_2 \left(\frac{-k_0}{2D_2} + C_3 \right)$$

$$C_1 = \frac{D_2}{D_1} \left(C_3 - \frac{k_0}{2D_2} \right)$$

$$C_{A1}(x) = C_1 (x+L) + C_L$$

BC#4 eval

$$1 = \frac{C_1 L + C_L}{\frac{k_0}{2D_2} L^2 - C_3 + C_R}$$

$$\frac{k_0}{2D_2} L^2 - C_3 + C_R = C_1 L + C_L$$

$$C_1 = \frac{\frac{k_0}{2D_2} L^2 - C_3 + C_R - C_L}{L}$$

$$\frac{\frac{k_0}{2D_2} L^2 - C_3 + C_R - C_L}{L} = \frac{D_2}{D_1} \left(C_3 - \frac{k_0}{2D_2} \right)$$

$$\frac{k_0 L^2}{2D_2} - C_3 + C_R - C_L = \frac{D_2 L}{D_1} C_3 - \frac{k_0 L}{2D_1}$$

$$C_3 = \frac{\frac{k_0 L^2}{2D_2} + (C_R - C_L) + \frac{k_0 L}{2D_1}}{\frac{D_2 L}{D_1} + 1}$$

$$C_1 = \frac{\frac{k_0}{2D_2} L^2 - C_3 + C_R - C_L}{L}$$

$$= \frac{\frac{k_0}{2D_2} L^2 - \frac{k_0 L^2}{2D_2} - (C_R - C_L) - \frac{k_0 L}{2D_1} + C_R - C_L}{\frac{D_2 L}{D_1} + 1}$$

$$= \frac{k_0 L}{2D_2} -$$